

Referee report on the paper “Trapping analysis of a magnetized electron by a circularly polarized electromagnetic wave in static electric field”

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The manuscript extends the relativistic pseudo-potential framework of Bellan (Phys. Plasmas, 2013) for the interaction of a magnetized electron with a circularly polarized electromagnetic wave to include a static electric field E_0 parallel to the background magnetic field. The authors derive an effective second-order equation for $\xi_z = 1 + \alpha\gamma'\beta'_z$, identify a trapping regime under anomalous Doppler resonance, and compare numerically the asymptotic energy partition between parallel and perpendicular kinetic energies with a ratio attributed to “quantum theory”. The topic is of interest and the extension is reasonable, but several issues require attention before publication. The trapping threshold quoted in the abstract (10^2) and in the body (10^6) differ by four orders of magnitude. The pseudo-potential of Eq. (49) is not a function of ξ_z alone. The “quantum” benchmark of Eq. (64) is the classical Manley–Rowe relation. The energy partition reported in Fig. 5 does not visibly close within the simulated single-particle setup. The substantial existing literature on anomalous-Doppler trapping is not cited. Finally, the chosen parameters do not support the application claims (runaway mitigation, plasma heating). Major revision is recommended.

1 Major comments

1.1 Discrepancy between the trapping threshold in the abstract and in the body

The abstract states that “effective trapping requires the wave electric field amplitude to exceed the static electric field by a factor of approximately 10^2 ”. The body (end of Sec. III, same parameters) gives $E_{\perp c}/|E_0| \approx 10^6$. Appendix B uses $E_w/|E_0| = 10^4$ (passing) and 10^8 (trapped), consistent with a threshold near 10^6 but bracketing it across four orders of magnitude. The abstract must be corrected, and the scan in Appendix B should include intermediate points (e.g. 10^5 , $10^{5.5}$, 10^6 , $10^{6.5}$, 10^7) to actually verify the predicted threshold.

1.2 The “pseudo-potential” is not a function of ξ_z alone

Eq. (49) contains a term $-\int \xi_z (\varsigma E_0/cB_0) \int \xi_z d\tau d\xi$ that depends explicitly on the history of the trajectory through the integral $\int \xi_z d\tau$. Consequently $W_0 = \frac{1}{2}(d\xi_z/d\tau)^2 + \psi(\xi_z)$ is *not* a conserved quantity in the Hamiltonian sense, and the graphical interpretation in Figs. 1(f) and 2(f) — where W_0 appears as a horizontal red dashed bound in the well — is only approximately valid. The authors acknowledge in passing that “ ψ also depends on t ” but do not quantify the validity domain. The derivation of Eq. (22) also invokes $\beta'_\perp \ll 1$, an approximation whose breakdown time should be stated explicitly given that the trapping regime is one in which $\beta'_\perp$ grows continuously. Please (i) derive an explicit criterion (in terms of $\varsigma E_0/cB_0$, integration time, and bounce frequency) under which the pseudo-potential picture is meaningful, (ii) add a diagnostic plot of $W_0(t)$ along the trajectories of Figs. 1 and 2 demonstrating approximate conservation or quantifying drift, and (iii) verify a posteriori that the dropped term in Eq. (18) remains subdominant over the time window shown.

1.3 Classical character of the energy-partition benchmark

The energy-partition ratio $\eta_\perp = 1/(1 + |\omega/\Omega|\gamma_c)$ in Eq. (64) is the standard Manley–Rowe relation for a resonant wave-particle interaction; it follows from the conservation of action variables and is

derivable, without any quantum input, from a Hamiltonian or single-particle classical analysis. The classical derivation is already contained in the very reference cited (Nezlin 1976), and the broader classical anomalous-Doppler literature (see comment 3.5) treats this energy partition without ever invoking $\hbar$. Labelling Eq. (64) as a “quantum theory” prediction, and presenting the agreement in Fig. 5 as a benchmark of “consistency between quantum theory and classical dynamics”, is therefore misleading: both sides reduce to the same classical conservation law and the agreement is essentially tautological. Please rephrase Section IV accordingly and reserve “quantum” language for the heuristic photon picture used in the discussion, where it serves a legitimate pedagogical purpose.

1.4 Energy balance in the trapped regime

The authors report $\eta_{\perp} \approx 0.448$ and $\eta_{\parallel} \approx 0$ during trapping (Fig. 5) and attribute the remaining $\sim 55\%$ to “stimulated wave emission” (Sec. V). The wave is prescribed and not back-reacted upon in the simulation, however, so no emission occurs in the calculation itself; the energy balance must close through the work done on the particle by the wave field, $W_{\text{wave}} = \int q \mathbf{E}_w \cdot \mathbf{v} dt$. Please compute this quantity along the simulated trajectory and verify that $\Delta W_{\parallel} + \Delta W_{\perp} = W_E + W_{\text{wave}}$, with $W_{\text{wave}} \approx -0.55 W_E$ in the trapped regime. This is the single-particle analogue of the stimulated-emission picture and would support the interpretation in Sec. V instead of merely asserting it.

1.5 Missing contact with the anomalous-Doppler trapping literature

Trapping of magnetized electrons by parallel-propagating circularly polarized waves under anomalous Doppler resonance is a long-studied problem, particularly in the space-plasma community (whistler-mode and EMIC interactions with energetic electrons in the radiation belts). The classical framework was developed by Karpman, Shklyar, Trakhtengerts and Nunn in the 1970s–80s, and substantially extended in recent years by Artemyev, Vasiliev, Mourenas, Bortnik and others, who describe trapping conditions, trapping fractions, energy and pitch-angle evolution during phase-trapped resonance, and the effect of inhomogeneities. The present paper’s actual contribution — the role of a parallel static E_0 in producing continuous $E_0 \rightarrow \varepsilon_{\perp}$ transfer in the trapped regime — should be explicitly positioned against this background. In particular, the trapping width of Eq. (49) should be compared with the standard $\sim \sqrt{\Omega \kappa'}$ result in the $E_0 \rightarrow 0$ limit, both to validate the reduction and to delineate what is new.

1.6 Parameter regime and the application claims

The simulations are performed at $\gamma_0 = 1.12$, $\beta_z \approx 0.45$, i.e. a mildly relativistic electron far from the runaway regime ($\gamma \sim 10\text{--}100$, $\beta_{\parallel} \rightarrow 1$). The frequency $\omega = 1.1 |\Omega|$ at refractive index $n = 4$ is also not in the whistler band ($\omega \ll |\Omega|$) referenced in [25] for runaway control. The applications proposed in the abstract and Sec. V (runaway mitigation, plasma heating) are therefore not directly supported by the parameters used. The heating claim in particular cannot follow from a single-particle calculation: a heating rate requires either a distribution-averaged argument or a self-consistent wave-particle treatment with back-reaction, neither of which is performed. Please either repeat the analysis at runaway- and whistler-relevant parameters, or temper the claims in Sections I, V, and VI accordingly.

2 Minor comments

1. Reframe Appendix B as a verification of the pseudo-potential reduction (the VPA does solve the full equations of motion without approximation, so the comparison is legitimate) rather than as an independent benchmark of the underlying physics — the latter would require relaxing the prescribed-wave / single-particle setup. Improve the scan resolution as noted in comment 3.1.
2. Several equation references are inconsistent or precede the equation they cite. The text before Eq. (20) refers to “Eq. (29)” while invoking the equation of motion for $\beta'_{\perp}$; Eq. (29) is given several pages later and the intended reference appears to be Eq. (12). Please go through the manuscript and renumber consistently.
3. Eq. (14), $\xi_{\perp} = 1 + \alpha \gamma' \beta'_{\perp}$, is dimensionally and notationally unclear (scalar = scalar + scalar $\times$ vector?). Clarify whether $\xi_{\perp}$ is a vector or whether $\beta'_{\perp}$ here denotes its modulus.

4. Sign convention for Ω : the authors state $q = -e$ and $\Omega < 0$ but several intermediate expressions are written with Ω where $|\Omega|$ would be more transparent. Adopt one convention consistently throughout.
5. Captions of Figs. 1, 2, 6 should include the SI values of γ_0 , γ' and the corresponding electron kinetic energy in keV, to situate the regime considered.
6. Page 13: “tokamak plasma with frequencies near the upper-hybrid mode” is misleading since $\omega = 1.1 |\Omega|$ is not generally the upper-hybrid frequency. Clarify.
7. Replace the software-specific “ode45” (Sec. III) by a generic description (e.g. “a standard adaptive Runge–Kutta integrator”).
8. Sign of E_0 : $E_0 = -0.2$ V/m vs. the statement that E_0 is along $\hat{z}$ — clarify in figure captions.
9. Typos: “motion equation” $\rightarrow$ “equation of motion” (multiple places in Sec. II); “the parallel velocity is no longer increase” $\rightarrow$ “no longer increases” (Sec. III); period vs. comma p. 9 (“...shown in Fig. 1, For the static...”).